TMUA / PRACTICE PAPER

Set B Paper 2

Mathematical Reasoning 20 questions · 75 minutes

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Polynomials

1. For a real constant k, consider the equation

$$x^4 - 4x^2 + k = 0$$

Which of the following conditions on k is **necessary and sufficient** for the equation to have exactly two distinct real solutions?

A $k > 4$

B $k = 4$

C $k \leq 0$

D $0 < k < 4$

E $k = 0$

F $k < 0$ or $k = 4$

G $k < 0$

Number

2. Let n be an integer. Consider the following conditions:

I: n is odd
II: n is a multiple of 4
III: n is a prime greater than 3

Which of these conditions is/are **sufficient** to guarantee that $n^3 - n$ is divisible by 12?

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II, and III

Polynomials

3. The curve C has equation $y = x^3 - 3x^2$

A tangent to C passes through the point $P(0, 1)$

The point at which the tangent touches C has x-coordinate t

Find the complete set of possible values of t

A $t = -\frac{1}{2}$ only

B $t = 1$ only

C $t = -\frac{1}{2}$ or $t = 1$

D $t = -1$ or $t = 1$

E $t = \frac{1}{2}$ or $t = 1$

F $t = -\frac{1}{2}$ or $t = \frac{3}{2}$

Algebra

4. A student is required to solve the equation

$$3\sqrt{x+1} = x - 1$$

Their attempt is as follows:

$$3\sqrt{x+1} = x - 1 \xrightarrow{\textbf{(I)}} 9(x+1) = (x-1)^2$$

$$9(x+1) = (x-1)^2 \xrightarrow{\textbf{(II)}} 9x + 9 = x^2 - 2x + 1$$

$$9x + 9 = x^2 - 2x + 1 \xrightarrow{\textbf{(III)}} x^2 - 11x - 8 = 0$$

$$x^2 - 11x - 8 = 0 \xrightarrow{\textbf{(IV)}} x = \frac{11 \pm 3\sqrt{17}}{2}$$

They then conclude that the solutions of the original equation are $x = \dfrac{11 + 3\sqrt{17}}{2}$ and $x = \dfrac{11 - 3\sqrt{17}}{2}$. Which one of the following is true?

A The solution is correct.

B Only one of $x = \dfrac{11 + 3\sqrt{17}}{2}$ and $x = \dfrac{11 - 3\sqrt{17}}{2}$ is correct and the error arises as a result of step (I).

C Only one of $x = \dfrac{11 + 3\sqrt{17}}{2}$ and $x = \dfrac{11 - 3\sqrt{17}}{2}$ is correct and the error arises as a result of step (II).

D Only one of $x = \dfrac{11 + 3\sqrt{17}}{2}$ and $x = \dfrac{11 - 3\sqrt{17}}{2}$ is correct and the error arises as a result of step (III).

E There is another value of x that satisfies the original equation and the error arises as a result of step (I).

F There is another value of x that satisfies the original equation and the error arises as a result of step (II).

G There is another value of x that satisfies the original equation and the error arises as a result of step (III).

Trigonometry

5. Let p be a real constant. Consider the equation

$$\sin x \cos x = p$$

on $0 \leq x < 2\pi$

Which of the following statements is/are true?

I: If $|p| < \frac{1}{2}$ then the equation has exactly 4 distinct solutions

II: If the equation has exactly 2 distinct solutions, then $|p| = \dfrac{1}{2}$

A None of them

B I only

C II only

D I and II

Logic

6. We say a sequence of real numbers x_n is *a Cauchy sequence* if

$$\text{for all } \varepsilon > 0 \text{ there exists a positive integer } N \in \mathbb{N} \text{ such that}$$
$$\text{for all } n, m \geq N \text{ we have } |x_n - x_m| < \varepsilon$$

Which one of the following statements is true if and only if x_n is **not** a Cauchy sequence?

A For every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $m, n \geq N$ with $|x_m - x_n| < \varepsilon$

B There exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ and for all $m, n \leq N$ we have $|x_m - x_n| \geq \varepsilon$

C For every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ and there exists $m, n \geq N$ with $|x_m - x_n| \geq \varepsilon$

D For every $N \in \mathbb{N}$ there exists $\varepsilon > 0$ such that for all $m, n \geq N$ we have $|x_m - x_n| \geq \varepsilon$

E There exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ there exist $m, n \geq N$ with $|x_m - x_n| \geq \varepsilon$

F There exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$ there exist $m, n \geq N$ with $|x_m - x_n| < \varepsilon$

G There exists $N \in \mathbb{N}$ such that for all $\varepsilon > 0$ and all $m, n \geq N$ we have $|x_m - x_n| \geq \varepsilon$

H There exists $\varepsilon > 0$ and there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$ we have $|x_m - x_n| \geq \varepsilon$

Integration

7. For a real constant k, define

$$A(k) = \int_{-1}^{2} |x - k| \, \mathrm{d}x$$

For which value of k is $A(k)$ minimised?

A -1

B $\frac{1}{2}$

C 1

D $\frac{3}{2}$

E 2

F 0

Proof & Error Analysis

8. Consider the following problem for an integer n:

$$\left(\frac{1}{3}\right)^{n+2} \geq \left(\frac{1}{27}\right)^{n-1}$$

A student produces the following argument:

I: $\left(\frac{1}{3}\right)^{n+2} \geq \left(\frac{1}{27}\right)^{n-1}$

II: $\log_{\frac{1}{3}}\left(\left(\frac{1}{3}\right)^{n+2}\right) \geq \log_{\frac{1}{3}}\left(\left(\frac{1}{27}\right)^{n-1}\right)$

III: $n+2 \geq (n-1)\log_{\frac{1}{3}}\left(\frac{1}{27}\right)$

IV: $n+2 \geq 3(n-1)$

V: $n+2 \geq 3n-3$

VI: $n \leq 2$

Which step (if any) in the argument is invalid?

A There are no invalid steps; the argument is correct.

B Only step **I** is invalid; the rest are correct.

C Only step **II** is invalid; the rest are correct.

D Only step **III** is invalid; the rest are correct.

E Only step **IV** is invalid; the rest are correct.

F Only step **V** is invalid; the rest are correct.

G Only step **VI** is invalid; the rest are correct.

Trigonometry

9. The equation

$$2\cos^2 x - k\sin x = 1$$

has n distinct solutions in the range $0 \leq x \leq 2\pi$

Consider the following statements:

I: $k = 0$ is **necessary** for $n = 2$
II: $k > 2$ **only if** $n = 0$

Which of the following statements is/are true?

A Neither of them

B I only

C II only

D I and II

Inequalities

10. A real number k has the property that

$$x^2 + kx + 1 \geq 0$$

for all real x satisfying $x^2 - 4x + 3 \leq 0$

Which of the following describes the set of all such values of k?

A $k \geq -2$

B $k \geq -\dfrac{10}{3}$

C $-2 \leq k \leq 2$

D $-4 \leq k \leq 2$

E $-4 \leq k \leq 4$

F $-6 \leq k \leq -2$

G $k \geq 0$

Functions & Graphs

11. Consider the curve defined by

$$(|x| - 4)^2 + (y - 1)^2 = 1$$

How many distinct straight lines passing through the origin are tangents to this curve?

A 0

B 1

C 2

D 3

E 4

F 5

Proof & Error Analysis

12. Let a and b be positive real numbers, neither of which is equal to 1.

A student attempts to prove the following claim:

Claim: If $\log_a b = \log_b a$ then $a = b$

The student writes the following argument:

Line 1: Let $t = \log_a b,$ so that $a^t = b$

Line 2: Since $b \neq 1,$ we have $t \neq 0$

Line 3: Raising both sides of $a^t = b$ to the power $\frac{1}{t}$ gives $a = b^{1/t}$

Line 4: Hence $\log_b a = \frac{1}{t},$ so the given condition implies $t = \frac{1}{t},$ and therefore $t^2 = 1$

Line 5: Since $a^t = b$ and both a and b are positive, we must have $t > 0$

Line 6: Hence $t = 1$

Line 7: Therefore $b = a^1 = a,$ which proves the claim.

Which line contains the first error?

A Line 1

B Line 2

C Line 3

D Line 4

E Line 5

F Line 6

G Line 7

H There is no error; the argument is correct.

Geometry

13. Let C be the circle $x^2 + y^2 = 9$

Let P be the point $(a, 0)$ where $a > 3$

The two tangents from P to C touch the circle at T_1 and T_2

What is the area of triangle PT_1T_2 in terms of a?

A $3\sqrt{a^2 - 9}$

B $\dfrac{3(a^2 - 9)^{3/2}}{a^2}$

C $\dfrac{9}{a}\sqrt{a^2 - 9}$

D $\dfrac{a}{3}\sqrt{a^2 - 9}$

E $\dfrac{a^2 - 9}{3}$

F $\dfrac{a^2 - 9}{a}$

G $\dfrac{9}{\sqrt{a^2 - 9}}$

Functions & Graphs

14. Let $f(x) = |x - 1|$ and $g(x) = |x + 1|$

Define

$$h(x) = f\big(g(x)\big)$$

How many integers x satisfy $h(x) < 1$?

A 0

B 1

C 2

D 3

E 4

F 5

Geometry

15. A cuboid has edge lengths a, b and c

The diagonals of its three different rectangular faces have lengths 5 cm, $\sqrt{153}$ cm and $4\sqrt{10}$ cm

What is the volume of the cuboid?

A 60 cm^3

B 144 cm^3

C 156 cm^3

D 180 cm^3

E 240 cm^3

F 624 cm^3

Proof & Error Analysis

16. A student attempts to prove the following claim.

Claim: For every real number k with $k > \frac{2}{3}$ the equation $x^3 - 3k^2x + k = 0$ has three distinct real solutions.

The student produces the following argument:

L1 Let $f(x) = x^3 - 3k^2x + k$, so $f'(x) = 3x^2 - 3k^2$, and $f'(x) = 0$ exactly when $x = -k$ or $x = k$,and these two values are different because $k \neq 0$

L2 We have $f'(x) > 0$ for $x < -k$ and for $x > k$, and $f'(x) < 0$ for $-k < x < k$ so f has a local maximum at $x = -k$ and a local minimum at $x = k$

L3 The local maximum value is $f(-k) = 2k^3 + k$, which is positive because $k > 0$

L4 The local minimum value is $f(k) = k - 2k^3 < 0$

L5 A cubic with positive x^3 coefficient whose local maximum value is positive and whose local minimum value is negative meets the x-axis at three different points, so the equation has three distinct real solutions

Which one of the following statements about this argument is correct?

A The first error is on line L1

B The first error is on line L2

C The first error is on line L3

D The first error is on line L4

E The first error is on line L5

F There is no error, and the claim is true.

G There is no error in any line, but the claim is false.

Polynomials

17. Consider the statement:

$$\text{For all real numbers } x, \textbf{ if } x^2 + px + 1 \leq 0 \textbf{ then } x \leq 2 \qquad (*)$$

What is the complete set of real values of p for which $(*)$ is true?

A no real values of p

B all real values of p

C $p \geq 0$

D $p < 0$

E $p \geq -\frac{5}{2}$

F $p < -4$ **or** $p > 0$

Sequences & Series

18. A sequence x_n is defined by $x_1 = 1$ and

$$x_{n+1} = \frac{1}{3}x_n + 2$$

for $n \geq 1$.

Let S be the sum to infinity of the series

$$\sum_{n=1}^{\infty} (x_{n+1} - x_n)$$

What is the value of S?

A $-\frac{3}{2}$

B -1

C 2

D $\frac{9}{2}$

E The series does not converge.

Number

19. The highest common factor of two positive integers is 18, and their lowest common multiple is 3780.

What is the smallest possible value of the sum of the two integers?

A 504

B 522

C 540

D 558

E 666

F 738

Logic

20. Consider the statement about a real number k:

$$\text{For all real numbers } x \text{ there exists a real number } y \text{ such that } (x-y)^2 + k(x+y) > 0 \qquad (*)$$

For which values of k is $(*)$ true?

A for no real values of k

B for all real values of k

C for all real k with $k > 0$

D for all real k with $k < 0$

E for all real k with $k \neq 0$

F for exactly one real value of k

G for all real k with $k \geq 0$

H for all real k with $k \leq 0$